

To measure the performance of an optimized compiler, I would evaluate metrics like execution time, CPU utilization, and memory consumption using benchmark programs. For instance, on my system with a 13th Gen Intel Core i7-13800H processor (14 cores, 20 threads, base clock around 1.8 GHz, turbo up to 5.0 GHz) and 32 GB of RAM, I would test how long a program takes to compile and execute before and after optimization using tools like time or performance profilers. A significant performance gain would be at least a 15-25% reduction in execution time, for example, decreasing from 10 seconds to around 7.5-8.5 seconds or a 10-20% reduction in memory use (e.g., 16 GB down to 13 GB). These improvements would indicate that the compiler's optimizations are efficiently utilizing the CPU's multi-core architecture and memory hierarchy without sacrificing program correctness.